area, then the flux through each turn will be the same. Then total emf induced in the coil is given by

$$\varepsilon = -N\frac{d\Phi_B}{dt} = -\frac{d(N\Phi_B)}{dt} \quad (4.3)$$

Here, $N\Phi_B$ is called flux linkage, defined as the product of number of turns N of the coil and the magnetic flux linking each turn of the coil Φ_B.

EXAMPLE 4.3

A cylindrical bar magnet is kept along the axis of a circular solenoid. If the magnet is rotated about its axis, find out whether an electric current is induced in the coil.

Solution

The magnetic field of a cylindrical magnet is symmetrical about its axis. As the magnet is rotated along the axis of the solenoid, there is no induced current in the solenoid because the flux linked with the solenoid does not change due to the rotation of the magnet.

EXAMPLE 4.4

A closed coil of 40 turns and of area 200 cm^2, is rotated in a magnetic field of flux density 2 Wb m^{-2}. It rotates from a position where its plane makes an angle of 30° with the field to a position perpendicular to the field in a time 0.2 sec. Find the magnitude of the emf induced in the coil due to its rotation.

Solution

N = 40 turns; B = 2 Wb m^{-2}
A = 200 cm^2 = 200 × 10^{-4} m^2;

Initial flux, $\Phi_i = BA\cos\theta$

$= 2\times200\times10^{-4}\times\cos60^\circ$

since $\theta = 90^\circ - 30^\circ = 60^\circ$

$\Phi_i = 2\times10^{-2}Wb$

Final flux, $\Phi_f = BA\cos\theta$

$= 2\times200\times10^{-4}\times\cos0^\circ$ since $\theta = 0^\circ$

$\Phi_f = 4\times10^{-2}Wb$

Magnitude of the induced emf is

$$\varepsilon = N\frac{d\Phi_B}{dt} = \frac{40\times\left(4\times10^{-2} - 2\times10^{-2}\right)}{0.2} = 4V$$

EXAMPLE 4.5

A straight conducting wire is dropped horizontally from a certain height with its length along east – west direction. Will an emf be induced in it? Justify your answer.

Solution

Yes! An emf will be induced in the wire because it moves perpendicular to the horizontal component of Earth's magnetic field.

4.1.4 Lenz's law

A German physicist Heinrich Lenz performed his own experiments on electromagnetic induction and deduced a

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